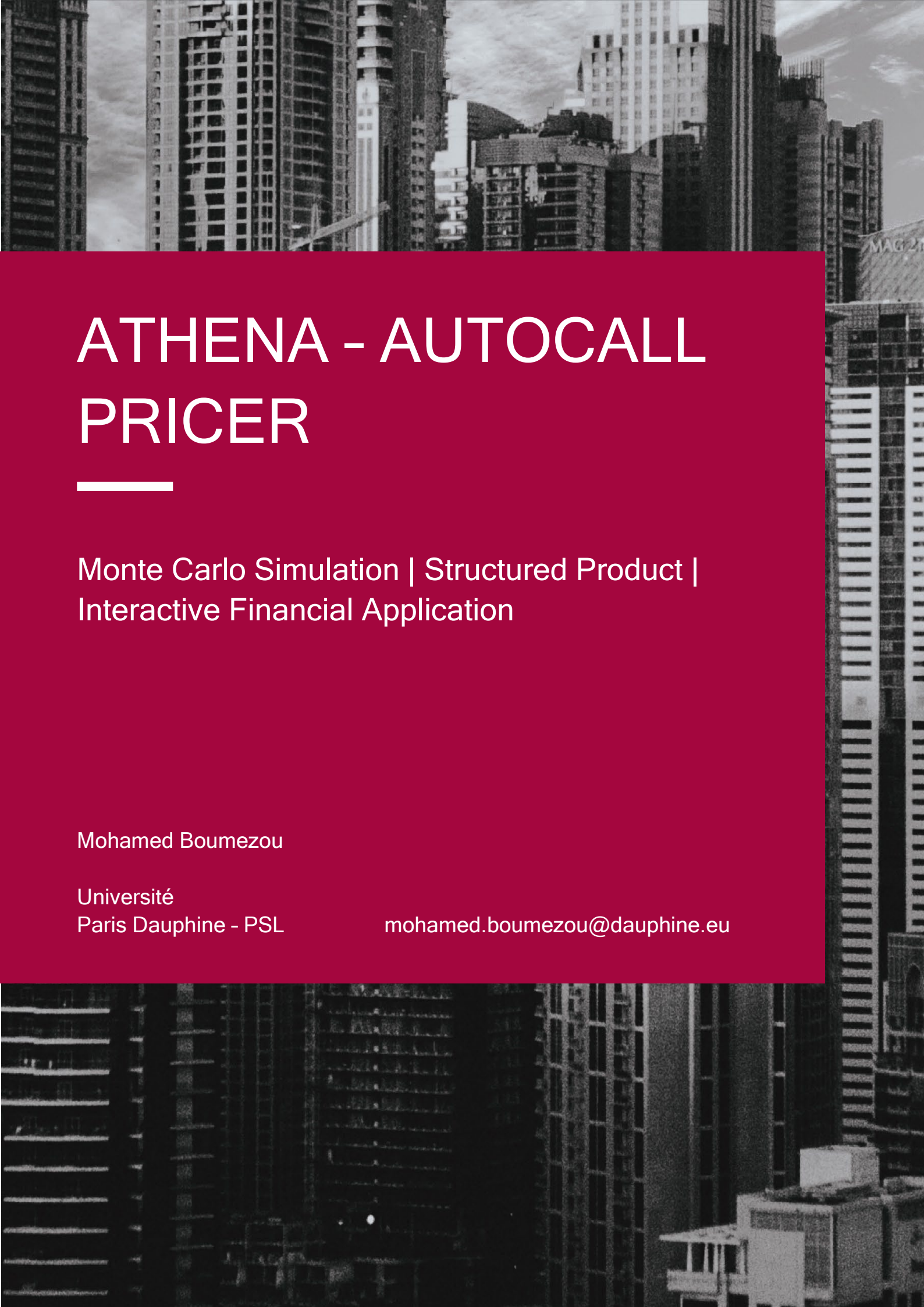




ATHENA - AUTOCALL PRICER

Monte Carlo Simulation | Structured Product |
Interactive Financial Application

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Presentation

Objective:

This document presents an interactive Python application for pricing a structured product of the **Autocall Athena** type, built with **Streamlit**.

The goal is to make financial modeling **accessible**, **interactive**, and **adaptable** to different market scenarios.

How the Autocall Athena works:

The Autocall Athena is a structured product whose return depends on the performance of an underlying asset (typically a stock or an index), with regular observation dates (generally annual).

It offers:

-  **Early redemption** if, on an observation date, the underlying is at or above its initial level → the investor receives the notional + coupon.
-  **Otherwise**, the product continues to the next date.
-  **At maturity:**
 - If underlying $\geq$ initial level → capital + coupon
 - If underlying $<$ initial but $\geq$ protection barrier → capital is returned
 - If underlying $<$ barrier → capital loss proportional to the decline

Mathematical modeling:

To simulate the behavior of the Autocall Athena, two key processes are modeled:

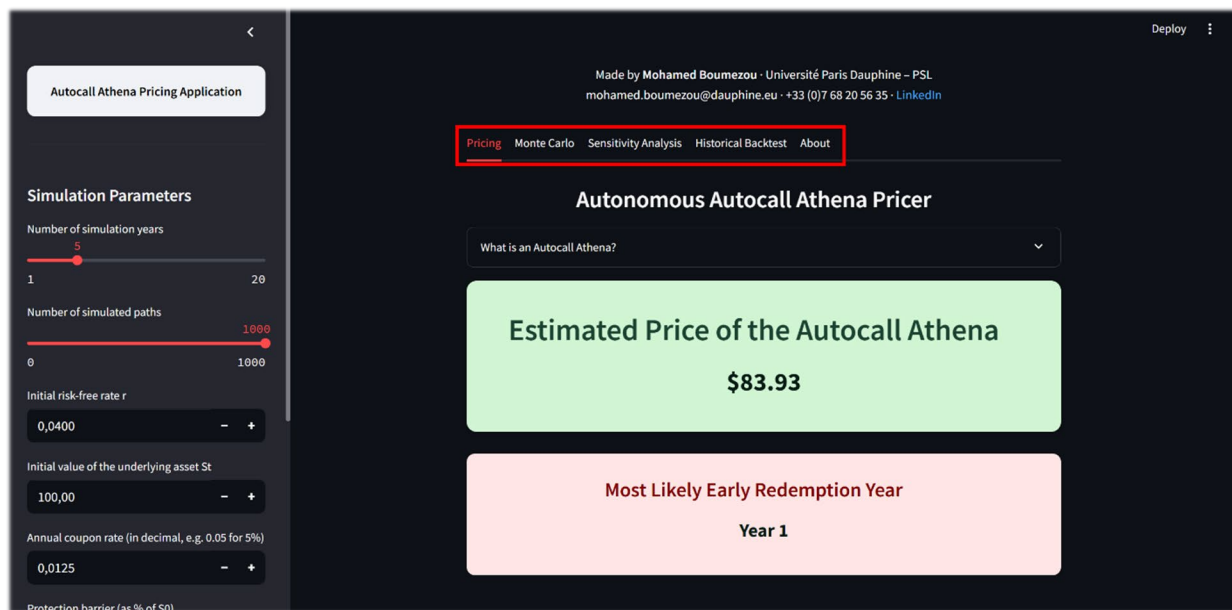
- The **risk-free rate**: modeled using a **CIR (Cox-Ingersoll-Ross)** stochastic process, widely used in interest rate modeling for its realistic and positive-only dynamics.
- The **underlying asset** (stock or index): its price evolves randomly according to a **local volatility** model, where volatility varies with both **time** and **price level**

Using The Pricer

1. Pricing Tab

The **Athena - Autocall Pricer** application allows users to simulate the price of a structured product of the Autocall type through an **intuitive and interactive interface**.

Users can define their own market assumptions and instantly visualize the results in real time. Below is an initial preview of the interface:



The application is organized into several tabs, each dedicated to a key aspect of the Autocall Athena modeling process:

- **Pricing** - Estimate the product's value and visualize redemption behavior
- **Monte Carlo** - Explore the underlying simulation models and adjust the volatility
- **Sensitivity Analysis** - Compute the main Greeks (Delta, Vega, Rho, Theta)
- **Historical Backtest** - Test the product on real financial data
- **About** - Learn more about the context and objectives behind the project

Customizing Simulation Parameters:

On the left side of the interface, users can customize key simulation inputs, including:

- Number of simulation years
- Number of simulated paths (Monte Carlo scenarios)
- Initial risk-free rate
- Initial value of the underlying asset
- Annual coupon rate
- Protection barrier level (as a percentage of the initial price)

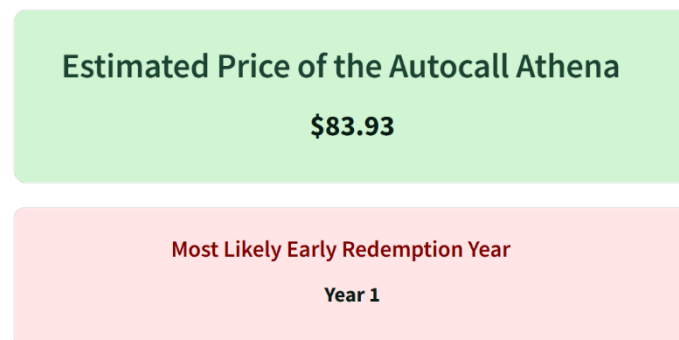
Note:

If you wish to adjust the **volatility**, you must go to the **Monte Carlo** tab and modify the local volatility function.

In this model, volatility is **not constant** – it is dynamically simulated and depends on both time and the asset price.

For optimal speed and performance, it is recommended to use **N < 300** simulations.

Simulation Results:

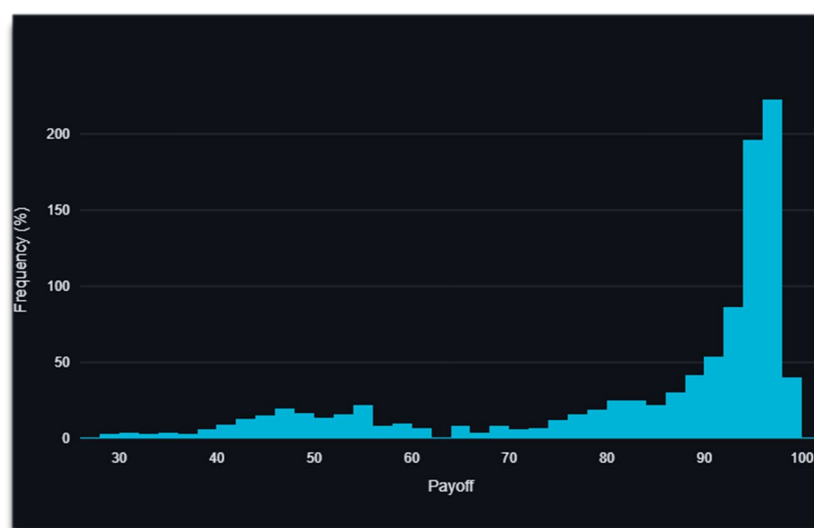


After running the Monte Carlo simulation, the application displays two key outputs:

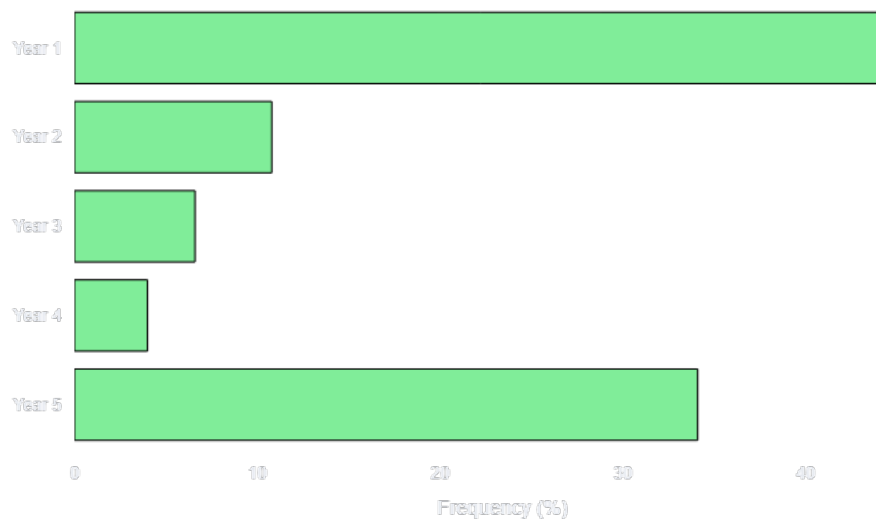
- **Estimated Price of the Autocall Athena:** This is the average payoff across all simulated paths, expressed in dollars. It reflects the expected value of the product under the user's assumptions.
- **Most Likely Early Redemption Year:** Based on the distribution of call events, the application identifies the year in which the product is most likely to be redeemed early.

These metrics provide quick and intuitive insights into the behavior and performance of the structured product.

Distribution graphs in Pricing Tab:



This histogram shows how payoffs are distributed across all simulated market scenarios. It provides insight into the **probability and variability** of outcomes for the Autocall Athena under the user-defined parameters. A concentration around higher payoffs suggests favorable conditions where the product is likely to be called early. A wider spread or noticeable presence of lower payoffs reflects greater exposure to adverse market movements and potential capital losses. This distribution is essential to assess the **risk-return profile** of the product.



This horizontal bar chart displays the probability that the Autocall Athena will be redeemed early during each observation year. Each bar represents the **frequency** at which the product was called in a given year across all simulated scenarios.

A high probability in early years reflects favorable market conditions that trigger early redemption. Conversely, if later years dominate, the underlying may struggle to reach the call level, increasing the risk of continuing until maturity.

This view helps investors assess the **timing and likelihood of early redemption**, a central feature of autocallable structures.

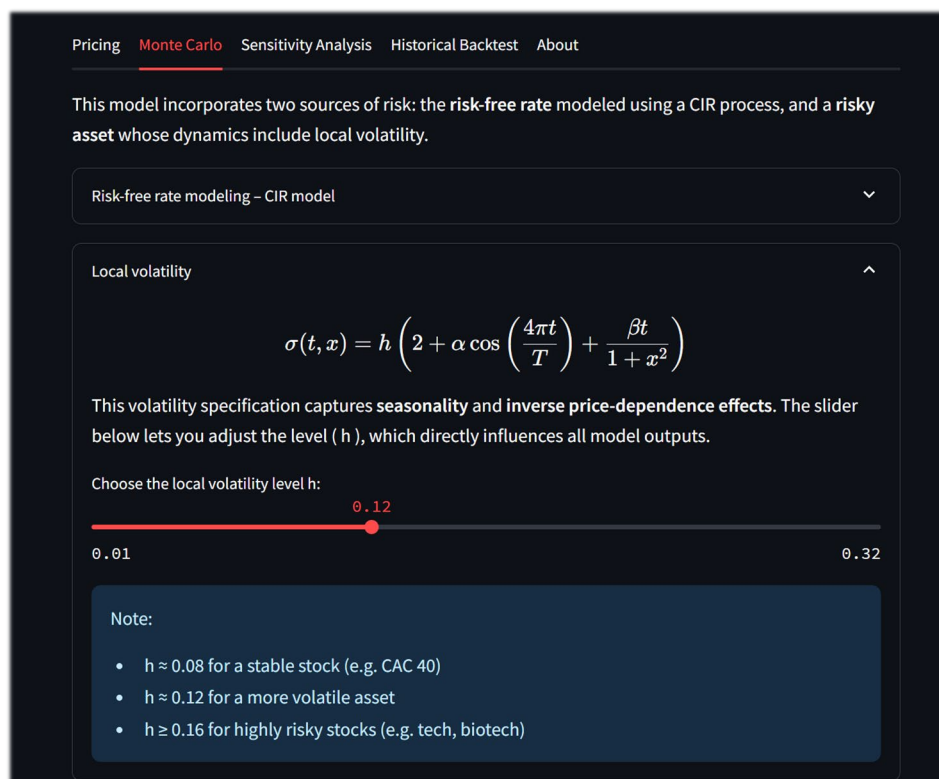
2. Monte Carlo Tab

Adjusting Local Volatility in Monte Carlo Tab:

To modify the local volatility parameter, open the **Monte Carlo** tab, then expand the **Local volatility** section.

There, you can view the exact formula used and adjust the level of h using the slider. This parameter controls the magnitude of volatility, which varies over time and depends on the asset price.

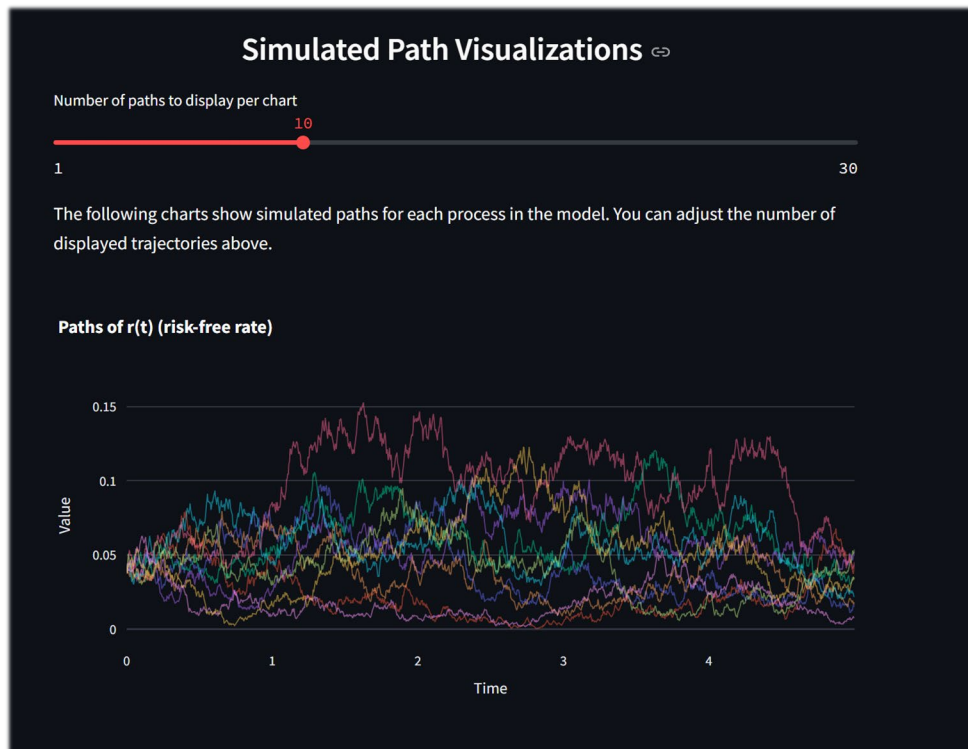
The note provided helps you select an appropriate value of h based on the underlying asset's risk profile.



Path visualization:

The application displays four interactive charts, each representing simulated paths for a key process in the model:

- risk-free interest rate
- risk-free asset
- risky asset (under local volatility)
- discounted value asset (final market value)



All charts are built using **Plotly**, which means:

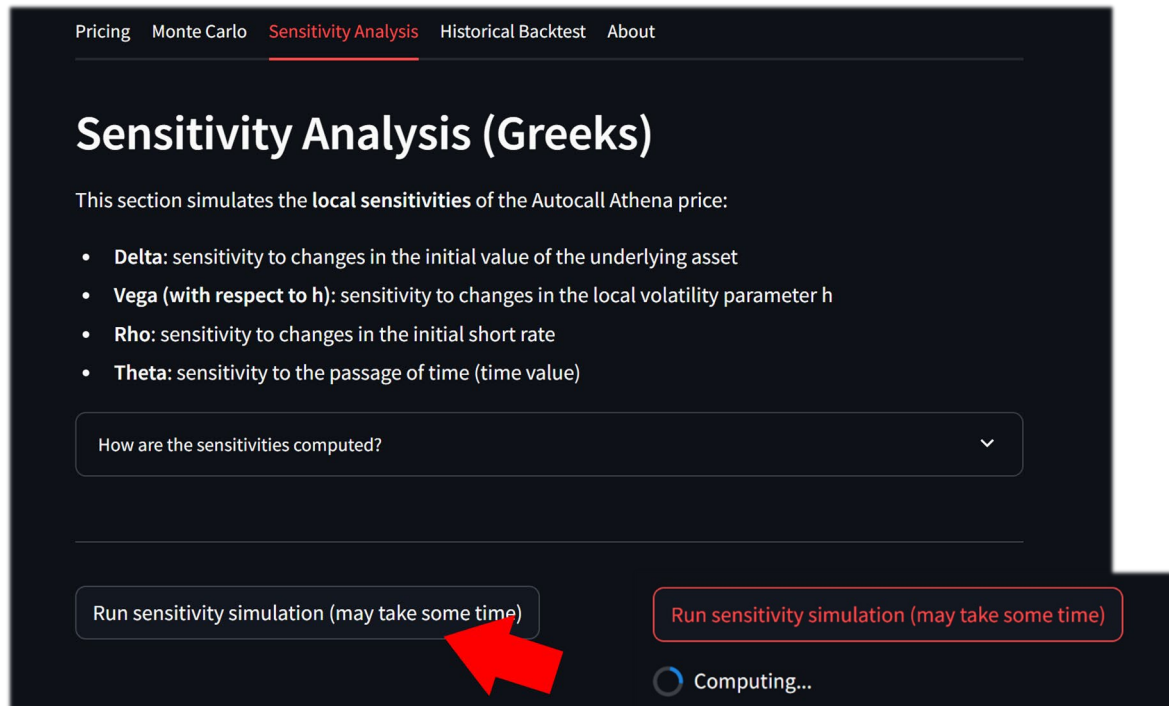
- They can be zoomed, panned, and hovered for detailed values
- Each chart can be expanded to fullscreen for better readability
- You can adjust the number of displayed paths dynamically using the slider

These visualizations help understand how each stochastic process evolves under different simulations.



3. Sensitivity Analysis Tab

Simulate the Greeks:



This section allows users to simulate the **local sensitivities** of the Autocall Athena price with respect to key market parameters:

- **Delta:** change in price with respect to the initial value of the underlying asset
- **Vega** (with respect to h): change in price with respect to the local volatility level
- **Rho:** sensitivity to changes in the initial risk-free rate
- **Theta:** impact of the passage of time (time decay)

These sensitivities are computed using the **finite difference method** and require multiple pricing simulations.

⚠ For this reason, the computation may take a few seconds-minutes depending on the number of paths selected (<300 is advised).

The results provide key insights into the **risk profile** of the structured product under various market conditions.

Sensitivity Results

Greek	Name	Value
Δ	Delta	0.7870
v	Vega (w.r.t h)	-9.5042
ρ	Rho	-9.9164
Θ	Theta	-2.0021

After the computation time, the application displays a table summarizing the estimated sensitivities—also known as **Greeks** of the Autocall Athena price.

4. Historical Backtest Tab

Backtest with Yahoo Finance API:

This backtest helps you assess the robustness of the product in a real scenario, and to compare theoretical vs. historical behavior.

Underlying ticker (e.g. TTE.PA)

TTE.PA

Start date

2021/01/01

End date

2025/01/01

Run backtest

This feature allows users to automatically retrieve historical data for a selected asset using its **ticker** symbol (e.g., TTE.PA) and a custom time period via Yahoo Finance. The application then **simulates the behavior of the Autocall Athena** over that historical window, **applying** the product's logic: early redemption conditions and capital protection rules.

The user can see:

- Whether the product would have been called early, and in which year
- The payoff that would have been achieved under actual market conditions

This allows for a direct comparison between theoretical pricing and real-world performance.

Conclusion:

This project was designed with the goal of making the modeling of a complex structured product accessible, by combining **financial rigor**, **pedagogical clarity**, and **interactivity**.

The application enables users to simulate, visualize, and interpret the behavior of an **Autocall Athena** under various market conditions, through a clear interface and customizable tools.

It reflects my desire not only to gain a deep understanding of financial products, but also to develop **practical tools** that can be useful to both **students** and **professionals**.

Contact:

For questions, feedback, or collaboration opportunities, feel free to reach out:

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